Assignment No	**7**
Title	**Spring Framework**
Objective	**Constructor Injection and Bean LifeCycle** **1. Design Student Application with beans Student and address inject dependencies using constructor injection**
Roll No	**MCA2542**

1: Design Student Application with beans Student and address inject dependencies using constructor injection

Source Code:

StudentImpl bean:

```
package edu.met.p1;
import org.springframework.context.*;
import org.springframework.context.support.*;

public class StudentImpl {
	private static ConfigurableApplicationContext ctx;
	public static void main(String[] args) {
		// TODO Auto-generated method stub
		ctx= new ClassPathXmlApplicationContext("AppCtx.xml");
		StudentBean s1=(StudentBean)ctx.getBean("Student");
		System.out.println(s1);
		ctx.close();
	}

}
```

StudentBean:

```
package edu.met.p1;
```

```
public class StudentBean {
	int rollno;
	String name;
	AddressBean address;
	public StudentBean(int rollno, String name, AddressBean address)
{
		super();
		this.rollno = rollno;
		this.name = name;
		this.address = address;
		System.out.println("Constructor called-Student");
	}
	//add init method
	public void init() throws Exception {
		System.out.println("Init Called");
	}
	//Add Destroy Method
	public void destroy() throws Exception{
		System.out.println("Destroy Called");
	}
	//Add toString
	@Override
	public String toString() {
		return "StudentBean [rollno=" + rollno + ", name=" + name
+ ", address=" + address + "]";
	}

}
```

AddressBean:

```
package edu.met.p1;

public class AddressBean {
	String street;
```

```
	String city;
	String state;
	public AddressBean(String street, String city, String state) {
		super();
		this.street = street;
		this.city = city;
		this.state = state;
		System.out.println("Constructor Called-Address");
	}
	@Override
	public String toString() {
		return "AddressBean [street=" + street + ", city=" + city + ",
state=" + state + "]";
	}


}
```

AppCtx:

```
<?xml version="1.0" encoding="UTF-8"?>
<beans
    xmlns="http://www.springframework.org/schema/beans"
    xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
    xmlns:p="http://www.springframework.org/schema/p"
    xsi:schemaLocation="http://www.springframework.org/schema/beans
             http://www.springframework.org/schema/beans/spring-
beans-3.0.xsd">
              <bean id="Address" class="edu.met.p1.AddressBean">
                <constructor-arg index="0" value="5th
Street"></constructor-arg>
                <constructor-arg index="1"
value="Mumbai"></constructor-arg>
                <constructor-arg index="2"
value="Maharashtra"></constructor-arg>
              </bean>
              <bean id="Student" class="edu.met.p1.StudentBean"
init-method="init" destroy-method="destroy">
                <constructor-arg index="0"
value="101"></constructor-arg>
                <constructor-arg index="1"
value="Nitin"></constructor-arg>
```

```
            <constructor-arg index="2">
                <ref bean="Address"/>
            </constructor-arg>
        </bean>
</beans>
```

Output:

```
Constructor Called-Address
Constructor called-Student
Init Called
StudentBean [rollno=101, name=Nitin, address=AddressBean [street=5th Street, city=Mumbai, state=Maharashtra]]
Destroy Called
```